

Axial Rod Stretching

Companion to the interactive simulator at <https://aimlab.uk/sims/axial-stretching.html>

Setup and assumptions

A straight rod of circular cross-section (diameter d) and length L is fixed at $x = 0$ and pulled by an axial force F at $x = L$. The material is linear elastic with Young's modulus E ; strains are small. Let $u(x)$ be the axial displacement of the cross-section at x .

Equilibrium

Cut the rod at any x : axial equilibrium of the free piece requires the internal normal force to equal the applied load, $N(x) = F$, at every section. Assuming the stress is uniform over the cross-section,

$$\sigma = \frac{N}{A} = \frac{F}{A}, \quad A = \frac{\pi d^2}{4}. \quad (1)$$

Kinematics and constitutive law

The axial strain is the displacement gradient, and Hooke's law relates it to stress:

$$\varepsilon = \frac{du}{dx}, \quad \sigma = E \varepsilon. \quad (2)$$

Governing equation

Combining the three ingredients, $N = EA u'$, and equilibrium with no distributed axial load gives $dN/dx = 0$, i.e.

$$EA \frac{d^2 u}{dx^2} = 0, \quad u(0) = 0, \quad EA u'(L) = F. \quad (3)$$

This is the axial analogue of the fourth-order bending equation — here the problem is second order. Integrating twice: $u = C_1 x + C_2$; the boundary conditions give $C_2 = 0$ and $C_1 = F/EA$.

Solution

$$\boxed{u(x) = \frac{Fx}{EA}} \quad \Rightarrow \quad \varepsilon = \frac{F}{EA} = \frac{\sigma}{E}, \quad \delta = u(L) = \frac{FL}{EA}. \quad (4)$$

Strain and stress are uniform along the rod; the elongation is linear in load and length, and inversely proportional to stiffness and area.

Worked example

$E = 10$ MPa (a soft elastomer), $d = 10$ mm, $L = 500$ mm, $F = 50$ N:

$$A = \frac{\pi \times 10^2}{4} = 78.5 \text{ mm}^2, \quad \sigma = \frac{50}{78.5 \times 10^{-6}} = 0.637 \text{ MPa}, \quad \varepsilon = 6.37\%, \quad \delta = 31.8 \text{ mm}. \quad (5)$$

Set the simulator sliders to these values and compare. Then switch E to steel (200 GPa): the same load produces $\delta = 1.6 \text{ } \mu\text{m}$ — the drawing does not visibly move, and that is the point. Beyond roughly 10% strain the small-strain, linear-elastic assumptions degrade.